

# Oluwatobi Melvyn Mayungbo

Software Engineer, Payments and Risk Systems

melvyn.t@gmail.com | [LinkedIn: linkedin.com/in/oluwatobi-mayungbo-3a567026b](https://www.linkedin.com/in/oluwatobi-mayungbo-3a567026b) | [GitHub: github.com/MelvTheGoat](https://github.com/MelvTheGoat)

## SUMMARY

---

Software engineer with a B.Sc. in Statistics who builds safe tools for AI and payments. Built an MCP server that checks crypto wallets and contracts for risk using only read-only calls, a double-entry ledger API on PostgreSQL with 138 tests, and a reconciliation system with a React review screen.

## SKILLS

---

**Engineering and MLOps:** Python, FastAPI, PostgreSQL, React, Docker, Pytest, CI/CD (GitHub Actions), SQL, DuckDB, Redis, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, Git, Streamlit

**Payments and Risk:** Web3 Fraud Risk Scoring, Double-Entry Ledgers and Idempotency, Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Regulatory Compliance (CBN/NDPA), Demand Forecasting

**AI and LLM Systems:** Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Hugging Face, ONNX Runtime

**Machine Learning:** PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

**Statistics and Causal Inference:** Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Time-Series Backtesting, Logistic Regression, WOE Scorecards

## PROJECTS

---

### web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

*Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions*

- Built an MCP server that gives AI assistants (Claude Desktop, Cursor) six read-only tools to check a crypto wallet, token or smart contract for risk before anyone interacts with it, returning a 0 to 100 risk score with a reason for every point, across Ethereum, Base, Arbitrum One, Polygon PoS and BNB Chain.
- Detects honeypot tokens, rug pull signs, hidden owner powers (mint, blacklist, pause), buy and sell tax, holder concentration, unlocked liquidity, and fund links to mixers, sanctioned wallets, exploiters and phishing addresses over 1 or 2 hops.
- Built a resilient HTTP layer with caching, per-source rate limits and retries with exponential backoff and jitter; one failing source never breaks an investigation, and missing data lowers confidence rather than the score.

### Production-Grade Double-Entry Ledger API | [GitHub](#)

*Tech: Python 3.12, FastAPI, PostgreSQL 16, Redis 7, Alembic, Docker*

- Built a double-entry ledger and payments API where correctness under concurrency, retries, partial failure and replay is enforced in the database: integer minor units, every transaction summing to zero through a deferred constraint trigger, and an append-only ledger where corrections are reversals.
- Wrote 138 tests against real PostgreSQL and Redis, including a concurrency test where exactly half of N simultaneous transfers succeed and no balance goes negative.

### Reckon: Payment Reconciliation and Review System | [GitHub](#) | [Live App](#)

*Tech: Python, PyTorch, DuckDB, FastAPI, React, Docker, Railway, PostgreSQL*

- Built and deployed a payment reconciliation system processing multi-source transaction feeds to detect uncollected revenue, duplicate settlements and payout discrepancies, combining deterministic matching rules with ML-assisted discrepancy resolution.
- Designed a human-in-the-loop review interface in React showing match confidence scores, audit trails and suggested resolution actions to streamline exception handling.

## EXPERIENCE

---

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
  - Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.
- Internal Auditor, NISER** May 2024 to July 2024
- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

**EDUCATION**

---

- B.Sc. Statistics, University of Ibadan** January 2021 to March 2025
- Professional Diploma in Artificial Intelligence, SQI College of ICT** July 2025 to Present

**CERTIFICATIONS**

---

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action